

Linear Programming

Table of contents

1	Day 1: Foundations of Linear Programming	3
1.1	Formulating Linear Programming Problems	3
1.1.1	A Practical Formulation Sequence	3
1.2	Farm LP Example	3
1.2.1	Scalar form	4
1.2.2	Matrix (compact) form	4
1.3	Properties of LP	4
1.4	Assumptions of Linear Programming	4
1.5	Graphical Solution Method (Two Variables)	4
1.5.1	Key Geometry Terms	4
1.6	Day 1 Summary	5
2	Day 2: Solving Linear Programs	7
2.1	Graphical solution method (2 variables)	7
2.1.1	Carbon abatement problem: algebraic solution	7
2.2	Simplex Method	8
2.2.1	Why We Need It	8
2.2.2	Core Idea	8
2.2.3	One pivot	8
2.2.4	Intuition	9
2.2.5	Optional Reference for Simplex Mechanics	9
2.2.6	Excel Solver	9
2.3	Day 2 Summary	9
3	Day 3: Sensitivity Analysis	10
3.1	Review	10
3.2	Motivation	10
3.3	What Is Sensitivity Analysis?	10
3.4	Walk-Through: Objective-Coefficient Sensitivity	10
3.4.1	Step 1: Start at the baseline optimum	10
3.4.2	Step 2: Write the relevant slopes	10
3.4.3	Step 3: Range for wheat's unit profit	11
3.4.4	Step 4: Range for corn's unit profit	11
3.4.5	Step 5: Separate activity levels from the objective value	12
3.5	Walk-Through: Resource Endowment (RHS) Sensitivity	12
3.5.1	Step 1: Identify resource use at the baseline optimum	12
3.5.2	Step 2: Define the shadow price	12
3.5.3	Step 3: Increase the binding labor endowment by one hour	12
3.5.4	Step 4: Check the nonbinding resources	13
3.5.5	Step 5: Derive the allowable labor range	13
3.5.6	Step 6: Derive the ranges for land and fertilizer	14
3.5.7	Step 7: Use the ranges for what-if questions	14

3.6	Solver Connection: Reduced Costs	14
3.7	Technical Coefficient Changes	15
3.8	Wheat–Corn Example (Excel Solver Report)	15
3.8.1	Excel Top Panel: Variable Cells	15
3.8.2	Excel Bottom Panel: Constraints	15
3.9	Economic Interpretation	16
3.10	Summary	16
4	Solve an LP in R	16
5	Duality in Linear Programming	17
5.1	Review	17
5.2	Wheat–Corn primal and its dual	17
5.2.1	An alternative to farming: offer the resources to someone else	17
5.2.2	What makes an offer adequate?	17
5.2.3	Interpreting the optimal offers	18
5.2.4	Why corn is grown and wheat is not	19
5.3	From the farm example to the general form	19
5.3.1	Canonical primal–dual pair	19
5.3.2	Constructing the dual in this form	19
5.4	Duality theorems	20
5.4.1	Weak duality	20
5.4.2	Strong duality	21
5.4.3	Where does the gap come from?	21
5.4.4	Complementary slackness conditions	21
5.5	Summary	22

1 Day 1: Foundations of Linear Programming

What is linear programming (LP)? It is a **mathematical optimization framework** for making decisions under constraints.

1.1 Formulating Linear Programming Problems

Kaiser and Messer emphasize that formulation is the central task in LP: the analyst translates a verbal decision problem into a mathematical model. Begin by reading the problem carefully to understand the decision setting before trying to write equations. Then read it again to sort the information needed for the model.

1.1.1 A Practical Formulation Sequence

1. **Identify the objective.** What is the decision maker trying to maximize or minimize—for example, profit, cost, sales, or production?
2. **Identify the activities (decision variables).** What choices can the decision maker make to pursue the objective? Define the unit of measurement for each activity.
3. **Identify objective coefficients.** For each activity, determine its per-unit contribution to the objective and record its units.
4. **Identify resources or requirements.** List the resources controlled by the decision maker, their available amounts, and their units. In problems without fixed resources, identify the relevant right-hand-side requirements instead.
5. **Identify technical coefficients.** For every activity-resource pair, determine how much of the resource is used or supplied by one unit of the activity. Check the units carefully.
6. **Write the structural constraints.** Translate resource limits or minimum requirements into $\leq$, $\geq$, or $=$ constraints.
7. **Add variable restrictions and assemble the model.** Include nonnegativity (or other sign restrictions) and combine the objective and constraints into a complete scalar model. Descriptive constraint labels make the model easier to audit and interpret.

Throughout the process, check that every term within an equation has compatible units. Formulation improves through practice: solve many verbal problems, compare alternative specifications, and ask whether each equation faithfully represents the decision setting.

1.2 Farm LP Example

A farmer has 500 acres of land available and is deciding how to allocate it between wheat and corn. Each acre of wheat yields a profit of \$200, requires 3 hours of labor, and 4 units of fertilizer. Each acre of corn yields a profit of \$300, requires 4 hours of labor, and 3 units of fertilizer. The farm has at most 1,800 hours of labor available and 2,000 units of fertilizer.

Formulate this situation as a linear programming problem. Clearly define the decision variables, write down the objective function representing total profit, and specify the constraints that capture the land, labor, fertilizer, and nonnegativity restrictions.

Decision variables.

Let W = acres of wheat, C = acres of corn.

1.2.1 Scalar form

$$\begin{aligned} \max_{W, C} \quad & 200W + 300C \\ \text{s.t.} \quad & W + C \leq 500 \quad (\text{land}) \\ & 3W + 4C \leq 1800 \quad (\text{labor}) \\ & 4W + 3C \leq 2000 \quad (\text{fertilizer}) \\ & W, C \geq 0. \end{aligned}$$

1.2.2 Matrix (compact) form

$$\begin{aligned} \max_{x \in \mathbb{R}_{\geq 0}^2} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b, \end{aligned} \quad \text{with} \quad x = \begin{bmatrix} W \\ C \end{bmatrix}, \quad c = \begin{bmatrix} 200 \\ 300 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 500 \\ 1800 \\ 2000 \end{bmatrix}.$$

1.3 Properties of LP

- **Objective** is to either maximize or minimize
- **Constraints:** linear inequalities or equalities that limit the feasible choices (e.g., land, labor, fertilizer).
- All equations are **linear**
- **Decision variables:** the choices to be made (e.g., how many acres of wheat and corn to plant) are non-negative.

1.4 Assumptions of Linear Programming

The first three assumptions concern whether the model represents the decision problem appropriately; the remaining four state its mathematical and information requirements.

- **Objective-function appropriateness:** one stated objective is sufficient to rank feasible choices.
- **Decision-variable appropriateness:** all relevant choices are included and controlled by the decision maker.
- **Constraint appropriateness:** the constraints accurately represent the relevant limits, requirements, and units.
- **Proportionality:** each activity's per-unit contribution and resource use do not change with its level.
- **Additivity:** the contribution of one activity does not depend on the level of another activity.
- **Divisibility:** decision variables may take fractional values.
- **Certainty:** objective, technical, and right-hand-side coefficients are known fixed values.

When an assumption is unsuitable, the model may need an extension: multi-objective programming, integer programming, nonlinear programming, or stochastic programming.

1.5 Graphical Solution Method (Two Variables)

1.5.1 Key Geometry Terms

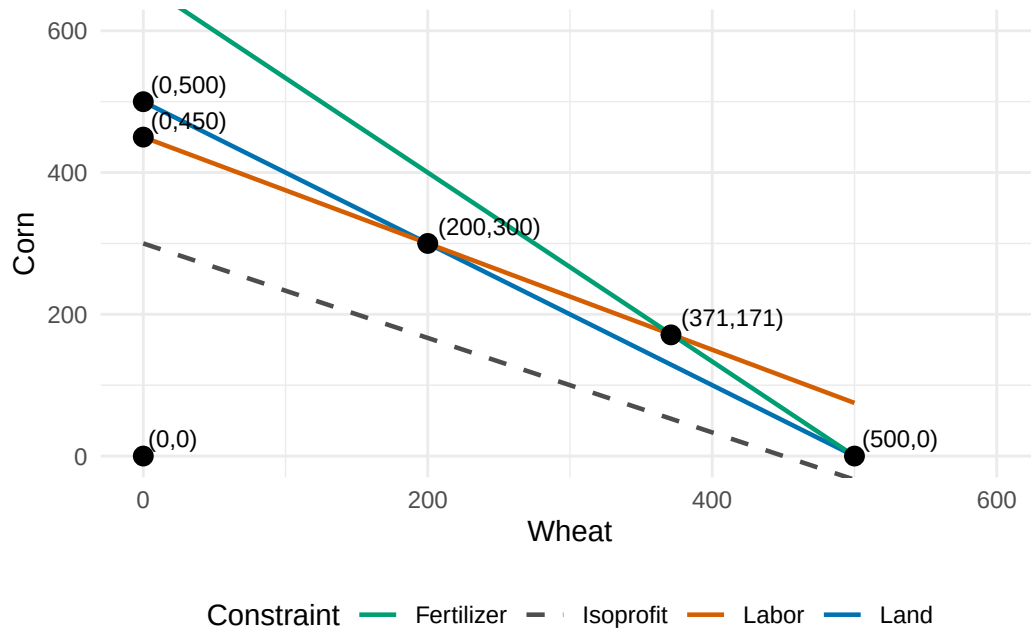
- A **feasible solution** satisfies every constraint; an **infeasible solution** violates at least one.
- The **feasible region** is the set of all feasible solutions.
- The feasible region of an LP is **convex**: the line segment joining any two feasible solutions is also feasible.
- An **extreme point** (or vertex) is a corner of the feasible region.
- A constraint is **binding** when it holds with equality at a solution; otherwise, it has **slack**.

Step 1. Draw the constraints.

- Land: $W + C \leq 500$
- Labor: $3W + 4C \leq 1800$
- Fertilizer: $4W + 3C \leq 2000$

- Nonnegativity: $W, C \geq 0$
- Plot based on endpoints. Set one variable to zero and devote all resources to the other variable.

Warning: package 'ggplot2' was built under R version 4.5.2



Step 2. Identify the feasible region.

- Intersection of all constraints in the (W, C) plane.
- Polygon bounded by lines.

Step 3. Plot the objective function.

- Profit = $200W + 300C$.
- Show isoprofit lines:
 - Suppose we plot \$90,000 profit. All wheat no corn, then all corn no wheat.
 - Lines of constant profit have slope $-\frac{200}{300} = -\frac{2}{3}$.

Step 4. Locate the optimum.

- Slide the isoprofit line outward until the last point of contact with the feasible region.
- Optimum is always at a **corner point** (fundamental theorem of LP).

Step 5. Verify algebraically.

Use the binding constraints to recover the exact solution, then check every constraint and calculate the objective value. In the farm example, the graphical optimum has $W = 0$ and the labor constraint binds:

$$4C = 1800 \Rightarrow C = 450.$$

The solution $(W, C) = (0, 450)$ satisfies land use ($450 \leq 500$) and fertilizer use ($1350 \leq 2000$), and it yields profit $200(0) + 300(450) = 135,000$.

1.6 Day 1 Summary

- LP is a framework for making decisions subject to constraints.

- Formulate an LP by identifying the objective, activities and their units, coefficients, resources or requirements, and variable restrictions.
- A standard LP combines a linear objective, linear structural constraints, and sign restrictions; its coefficients are summarized in c , A , and b .
- LP relies on assumptions about appropriate objectives, variables, and constraints, along with proportionality, additivity, divisibility, and certainty.
- In two-variable problems, graph the constraints to find the feasible region, use an isoprofit line to locate an extreme-point optimum, and verify the result algebraically.

2 Day 2: Solving Linear Programs

2.1 Graphical solution method (2 variables)

Vertex	Profit
$(W, C) = (0, 0)$	\$0
$(0, 450)$	\$135,000
$(200, 300)$	\$130,000
$(500, 0)$	\$100,000

- Why don't we need to check (371,171)? Because it is not a vertex of the feasible region.

Fundamental theorem of linear programming: If an LP has an optimal solution, at least one of them occurs at a vertex of the feasible region.

- Which constraints bind at the optimum?
- What parameters of the problem would need to change for wheat to enter the solution?

2.1.1 Carbon abatement problem: algebraic solution

Write the carbon-abatement problem in matrix form. Identify the decision variables, objective coefficients, technical coefficients, and right-hand-side requirements.

- The carbon-abatement model has three decision variables.

Show 3d plot of carbon-abatement feasible region.

- We can find a vertex by setting **three independent constraints** to equality.
- The candidate binding constraints can be emissions requirements or nonnegativity restrictions such as $B = 0$.

For example, set the nitrous-oxide requirement and $M = F = 0$ to equality:

$$5B = 75 \Rightarrow (B, M, F) = (15, 0, 0).$$

This candidate is feasible because it reduces carbon dioxide by $150 \geq 100$, methane by $30 \geq 25$, and nitrous oxide by 75. Its cost is $0.5(15) = \$7.5$ million.

To find another candidate, set the carbon-dioxide and nitrous-oxide requirements to equality and set $F = 0$:

$$\begin{aligned} 10B + 15M &= 100, \\ 5B + 12M &= 75. \end{aligned}$$

- Using Gaussian elimination, we can solve this system of equations. Doubling the second equation and subtracting the first gives $9M = 50$, so $M = 50/9$ and $B = 5/3$. This point is feasible because its methane reduction is $280/9 \approx 31.1 \geq 25$.

Repeating this process for all combinations of three binding constraints, and discarding any intersection that violates a remaining constraint, gives the feasible vertices below.

Binding constraints	(B, M, F)	Cost (million \$)
$\text{CO}_2, \text{N}_2\text{O}, B = 0$	$(0, 115/21, 5/7)$	8.93
$\text{CO}_2, \text{N}_2\text{O}, F = 0$	$(5/3, 50/9, 0)$	7.50

Binding constraints	(B, M, F)	Cost (million \$)
$\text{CO}_2, B = 0, F = 0$	$(0, 20/3, 0)$	8.00
$\text{N}_2\text{O}, B = 0, M = 0$	$(0, 0, 75/13)$	19.04
$\text{N}_2\text{O}, M = 0, F = 0$	$(15, 0, 0)$	7.50

The smallest cost is \$7.5 million. Because two vertices tie, every feasible point on the line segment between $(5/3, 50/9, 0)$ and $(15, 0, 0)$ is also optimal.

This is an example of **multiple optimal solutions**.

2.2 Simplex Method

2.2.1 Why We Need It

- The graphical method only works for **two or three variables**. Algebraic methods are tedious for large problems.
- Real problems may involve **hundreds or thousands** of variables.
- The simplex method provides a **systematic way** to move from vertex to vertex until the best one is found.

2.2.2 Core Idea

- Start at a **basic feasible solution (BFS)** — a corner point of the feasible region.
- At each step:
 1. Identify an **entering variable**: an activity whose introduction can improve the objective.
 2. Determine which current activity must decrease to keep the solution feasible; this is the **leaving variable**.
 3. **Pivot** to a neighboring BFS.
- Stop when no variable can improve the objective — this is optimal.

2.2.3 One pivot

Return to the wheat–corn model from Day 1. We can use it to see what the simplex terms mean without carrying out a long sequence of row operations:

$$\begin{aligned}
 \max \quad & 200W + 300C \\
 \text{s.t.} \quad & W + C \leq 500 \quad (\text{land}), \\
 & 3W + 4C \leq 1800 \quad (\text{labor}), \\
 & 4W + 3C \leq 2000 \quad (\text{fertilizer}), \\
 & W, C \geq 0.
 \end{aligned}$$

Add slack variables to write the resource constraints as equalities:

$$\begin{aligned}
 W + C + s_L &= 500, \\
 3W + 4C + s_K &= 1800, \\
 4W + 3C + s_F &= 2000, \\
 s_L, s_K, s_F &\geq 0.
 \end{aligned}$$

At the origin, $W = C = 0$, so $(s_L, s_K, s_F) = (500, 1800, 2000)$. This is a BFS: the slack variables form the initial **basis** and record the unused resources.

- At this point, a unit of corn raises profit by \$300, compared with \$200 for a unit of wheat. Choose C as the **entering variable**.

- Let $C = t$ while $W = 0$. The slack equations become $s_L = 500 - t$, $s_K = 1800 - 4t$, and $s_F = 2000 - 3t$. Feasibility requires

$$t \leq 500, \quad t \leq \frac{1800}{4} = 450, \quad t \leq \frac{2000}{3} \approx 666.7.$$

- Labor is reached first, so s_K is the **leaving variable**. This is the **ratio test**: the smallest nonnegative limit determines how far the entering activity can increase.
- The pivot moves us to the neighboring vertex $(W, C) = (0, 450)$, where $(s_L, s_K, s_F) = (500, 0, 650)$ and $z = \$135,000$. The basis changes from $\{s_L, s_K, s_F\}$ to $\{s_L, C, s_F\}$.

Can we improve profit by adding wheat?

- To remain on the binding labor constraint, one additional acre of wheat requires reducing corn by 3/4 acre (ratio of technical coefficients). The resulting change in profit is

$$200 - 300 \left(\frac{3}{4} \right) = -25.$$

Thus wheat has no profitable direction in which to enter, and the current BFS is optimal. The simplex algorithm performs this same check-and-move process in higher dimensions, using matrix operations rather than a graph.

2.2.4 Intuition

- Simplex is like **walking along the edges** of the feasible polygon.
- At each corner, ask: *“If I move along this edge, does profit go up?”*
- Continue until no edge yields improvement.
- The same logic applies in higher dimensions, even though we cannot draw the polytope.

2.2.5 Optional Reference for Simplex Mechanics

For a matrix-algebra treatment of the simplex algorithm, see McCarl and Spreen, Chapter III, especially Sections 3.2 and 3.4. The graphical method above supplies the intuition needed to interpret what the algorithm is doing.

2.2.6 Excel Solver

Excel implements the simplex method in the solver add-on. See [LP_land_alloc.xlsx](#)

2.3 Day 2 Summary

- If a bounded LP has an optimum, at least one optimal solution occurs at a **vertex** of its feasible region.
- **Simplex** generalizes the graphical corner-point search: it moves from one **basic feasible solution (BFS)** to a neighboring BFS until no feasible move improves the objective.
- Slack variables convert “ $\leq$ ” constraints to equalities, measure unused resources, and often provide a convenient initial basis.
- In each iteration, choose an **entering variable** with potential to improve the objective; the **ratio test** identifies the **leaving variable** and keeps the solution feasible; a **pivot** updates the basis.
- A different feasible starting vertex may change the path simplex takes, but not the optimal objective value.

3 Day 3: Sensitivity Analysis

3.1 Review

- LPs make 7 assumptions. What are a few?
- How do LPs algorithms solve? What is the fundamental theorem of LP?

3.2 Motivation

- LPs give an **optimal solution** and an **objective value**.
 - But: parameters (profits, resources, input requirements) are rarely known with certainty.
 - We need to know how **robust** the solution is.
-

3.3 What Is Sensitivity Analysis?

- Also called **post-optimality analysis**.
- Asks how changes in model parameters affect the optimal activities and objective value.
- It supports both **what-if analysis** and, when a price varies systematically, a **price-quantity schedule** such as an individual supply curve.
- Focus on three categories:
 1. Objective coefficients (π_j : unit profits, costs, or prices),
 2. RHS values (b_i : resource endowments or requirements), and
 3. Technical coefficients (a_{ij} : input-output relationships).

The objective-coefficient walkthrough below follows the sequence in Kaiser and Messer, Chapter 1, pp. 34–37, but applies every step to our wheat–corn farm model.

3.4 Walk-Through: Objective-Coefficient Sensitivity

Returning to the wheat–corn farm example, the LP is:

$$\begin{aligned} \max_{W,C} \quad & Z = 200W + 300C \\ \text{s.t.} \quad & W + C \leq 500 && (\text{land}) \\ & 3W + 4C \leq 1800 && (\text{labor}) \\ & 4W + 3C \leq 2000 && (\text{fertilizer}) \\ & W, C \geq 0. \end{aligned}$$

3.4.1 Step 1: Start at the baseline optimum

The baseline solution is $W^* = 0$, $C^* = 450$, and $Z^* = \$135,000$. Labor is binding; land and fertilizer have slack. The point also lies on the nonnegativity boundary $W = 0$.

Changing an objective coefficient changes the direction of an iso-profit line, but it does not move the feasible region. The question is therefore: how far can that line rotate before another corner becomes better than the corn-only corner?

3.4.2 Step 2: Write the relevant slopes

Use general unit-profit coefficients π_W and π_C :

$$\max Z = \pi_W W + \pi_C C.$$

Solving an iso-profit line for corn gives

$$C = \frac{Z}{\pi_C} - \frac{\pi_W}{\pi_C}W,$$

so its slope is $-\pi_W/\pi_C$.

The labor constraint, which is the feasible edge leaving the corn-only corner, is

$$C = \frac{b_{labor}}{a_{lc}} - \frac{a_{lw}}{a_{lc}}W = 450 - \frac{3}{4}W,$$

with slope $-3/4$.

For the corn-only corner to remain optimal, moving rightward along the labor boundary must not increase profit. Thus, the iso-profit line must be no steeper than that edge:

$$-\frac{\pi_W}{\pi_C} \geq -\frac{3}{4} \quad \Longleftrightarrow \quad \frac{\pi_W}{\pi_C} \leq \frac{3}{4}.$$

Unlike the reading's feed-dealer example, this corner has only one adjacent structural constraint edge; the other binding condition is $W = 0$. Therefore the coefficient ranges below have one finite endpoint rather than two.

3.4.3 Step 3: Range for wheat's unit profit

Hold corn's unit profit at $\pi_C = 300$. Then

$$\frac{\pi_W}{300} \leq \frac{3}{4} \quad \Rightarrow \quad \pi_W \leq 225.$$

So wheat's unit profit can rise from \$200 to \$225 per acre without changing the unique corn-only optimum. At $\pi_W = 225$, the labor edge has the same slope as the iso-profit line: every feasible mix on that edge from $(W, C) = (0, 450)$ to $(200, 300)$ is optimal. Above \$225, wheat enters the optimal plan.

There is no finite allowable decrease for wheat's objective coefficient in this example: making an activity that is already unused even less attractive does not displace corn.

3.4.4 Step 4: Range for corn's unit profit

Hold wheat's unit profit at $\pi_W = 200$. Then

$$\frac{200}{\pi_C} \leq \frac{3}{4} \quad \Rightarrow \quad \pi_C \geq 266.67.$$

Corn's unit profit can fall from \$300 to \$266.67 per acre without changing the unique corn-only optimum. At $\pi_C = 266.67$, the same alternative-optima edge appears; below that level, wheat enters the plan. There is no finite allowable increase for corn's unit profit, because a higher corn return reinforces the corn-only solution.

3.4.5 Step 5: Separate activity levels from the objective value

Within the range, the activity levels stay at $W = 0$ and $C = 450$, but profit changes with the coefficient. For example, if corn's unit profit rises to \$320, the plan remains corn-only while profit becomes

$$Z^* = 320(450) = \$144,000.$$

This is the central distinction in objective-coefficient ranging: the **optimal basis and activity levels** can remain unchanged even though the **optimal objective value** changes.

Discussion prompts

1. If wheat profit rises to \$220, should we re-solve the LP? Why or why not?
 2. Why is \$225 a threshold for alternative optima rather than a point at which a solver must choose a particular positive wheat acreage?
 3. If both crop prices change, why must we check the ratio π_W/π_C rather than apply two one-at-a-time ranges independently?
-

3.5 Walk-Through: Resource Endowment (RHS) Sensitivity

This walkthrough follows the resource-endowment method in Kaiser and Messer, Chapter 1, pp. 38–40, but applies it to the wheat–corn farm. Hold the objective coefficients and all other model parameters constant while changing one RHS value at a time.

3.5.1 Step 1: Identify resource use at the baseline optimum

The three resource endowments are

$$b_{\text{land}} = 500, \quad b_{\text{labor}} = 1800, \quad b_{\text{fert}} = 2000.$$

At $(W^*, C^*) = (0, 450)$, the farm uses 450 acres of land, 1,800 labor hours, and 1,350 units of fertilizer. Labor is binding; land has 50 acres of slack and fertilizer has 650 units of slack.

Question. If the farm obtains one additional unit of a resource, how much does optimal profit change?

3.5.2 Step 2: Define the shadow price

Following Kaiser and Messer, change resource i by a small amount, re-solve the model, and compare the new and original optimal values:

$$SP_i = \frac{Z^*(b_i + \Delta b_i) - Z^*(b_i)}{\Delta b_i}.$$

For a one-unit increase, $\Delta b_i = 1$, so the shadow price is simply the change in optimal profit. Because an LP value function is piecewise linear, this marginal value is exact for any RHS change that remains within the allowable range.

3.5.3 Step 3: Increase the binding labor endowment by one hour

Raise labor availability from 1,800 to 1,801 hours. If the current basis remains optimal, wheat stays at zero and the new labor constraint binds:

$$3W + 4C = 1801, \quad W = 0.$$

Therefore,

$$C' = \frac{1801}{4} = 450.25.$$

Check the remaining constraints:

$$450.25 \leq 500 \quad (\text{land}), \quad 3(450.25) = 1350.75 \leq 2000 \quad (\text{fertilizer}).$$

The new solution is feasible and its profit is

$$Z' = 300(450.25) = \$135,075.$$

Thus,

$$SP_{\text{labor}} = \frac{135,075 - 135,000}{1801 - 1800} = \$75 \text{ per labor hour}.$$

One more labor hour permits one-quarter acre of additional corn, worth $300(1/4) = 75$ dollars. This is the operation's marginal value of labor capacity before paying to acquire it.

3.5.4 Step 4: Check the nonbinding resources

Increasing land from 500 to 501 acres leaves the plan at $(0, 450)$: land slack rises from 50 to 51 acres, while profit remains \$135,000. Therefore, $SP_{\text{land}} = 0$ at the current endowment.

Likewise, increasing fertilizer from 2,000 to 2,001 units only raises fertilizer slack. The activity levels and profit do not change, so $SP_{\text{fert}} = 0$ at the current endowment.

A zero shadow price does not mean that a resource can never have value. It means that one additional unit has no value **at the current endowment and basis**.

3.5.5 Step 5: Derive the allowable labor range

Let labor availability be the variable RHS b_{labor} . As long as the corn-only basis remains feasible,

$$C = \frac{b_{\text{labor}}}{4}, \quad s_{\text{land}} = 500 - \frac{b_{\text{labor}}}{4}, \quad s_{\text{fert}} = 2000 - \frac{3b_{\text{labor}}}{4}.$$

All basic variables must remain nonnegative. Ask what would b_{labor} have to be for each basic variable to become zero:

$$\begin{aligned} C \geq 0 &\implies b_{\text{labor}} \geq 0, \\ s_{\text{land}} \geq 0 &\implies b_{\text{labor}} \leq 2000, \\ s_{\text{fert}} \geq 0 &\implies b_{\text{labor}} \leq 2666.67. \end{aligned}$$

The tightest conditions give

$$0 \leq b_{\text{labor}} \leq 2000.$$

Starting from 1,800 hours, the allowable decrease is 1,800 hours and the allowable increase is 200 hours. Within this range,

$$C^* = \frac{b_{\text{labor}}}{4}, \quad Z^* = 75b_{\text{labor}},$$

so the shadow price remains \$75 per hour. At 2,000 hours, land also becomes binding. Beyond that breakpoint, the farm cannot expand corn acreage, so the marginal value of more labor falls to zero.

3.5.6 Step 6: Derive the ranges for land and fertilizer

At the baseline solution, corn uses 450 acres and 1,350 fertilizer units. Therefore:

- The current basis remains feasible for $b_{\text{land}} \geq 450$. The allowable decrease is 50 acres, the allowable increase is unbounded, and the current shadow price is zero.
- The current basis remains feasible for $b_{\text{fert}} \geq 1350$. The allowable decrease is 650 units, the allowable increase is unbounded, and the current shadow price is zero.

The one-at-a-time RHS sensitivity results are:

Resource	Current RHS	Use	Slack	Shadow price	Allowable decrease	Allowable increase
Land	500	450	50	\$0/acre	50	∞
Labor	1,800	1,800	0	\$75/hour	1,800	200
Fertilizer	2,000	1,350	650	\$0/unit	650	∞

3.5.7 Step 7: Use the ranges for what-if questions

- **Add 100 labor hours.** This is within the allowable increase. Corn rises to 475 acres and profit rises by $75(100) = 7,500$ dollars.
- **Add 250 labor hours.** This exceeds the allowable increase. Applying \$75 to all 250 hours would incorrectly predict an \$18,750 gain. Re-solving gives 500 corn acres and \$150,000 profit, a gain of only \$15,000, because land becomes binding after the first 200 additional labor hours.
- **Lose 75 acres of land.** The first 50 acres reduce slack but not profit. The remaining 25-acre loss forces corn down to 425 acres, reducing profit by $300(25) = 7,500$ dollars.

At an allowable-range endpoint, interpret the shadow price as one-sided: the marginal value on one side of the kink can differ from the marginal value on the other.

Discussion prompts

1. Why does labor have a positive shadow price while land and fertilizer initially have zero shadow prices?
2. Why can activity levels change even though the current basis and shadow price remain valid?
3. What resource becomes binding after labor rises above 2,000 hours, and how does that alter the value of additional labor?

3.6 Solver Connection: Reduced Costs

- The graphical condition above says that wheat can enter only when its unit profit exceeds \$225. In the Excel maximization convention, wheat's reduced cost is therefore $200 - 225 = -25$.
- A nonbasic activity has a reduced cost of zero at the entry threshold; at the threshold, alternative optima can occur. Sign conventions differ across solvers, so always interpret the reported value using the solver's documentation.
- Changing an objective coefficient never changes primal feasibility. For a basic activity, the solver checks whether the implied dual prices and reduced costs remain feasible.

3.7 Technical Coefficient Changes

- What if technology or input requirements change?
- A change in a_{ij} changes the feasible region itself, so it is not covered by the usual one-at-a-time objective-coefficient or RHS ranges in a Solver sensitivity report.
- For a small change that preserves the current solution, a shadow-price approximation is

$$\Delta Z^* \approx -y_i^* x_j^* \Delta a_{ij},$$

where y_i^* is the shadow price of resource i and x_j^* is the activity level.

- Example: if corn's labor requirement rises from 4 to 4.02 hours per acre, the wheat–corn model predicts $\Delta Z^* \approx -75(450)(0.02) = -\675 . Re-solve for a material change.
- If an activity is currently zero, changing one of its technical coefficients has no local effect on the objective value, although a sufficiently large change can make it enter.

3.8 Wheat–Corn Example (Excel Solver Report)

3.8.1 Excel Top Panel: Variable Cells

Interpretation of columns:

- **Final Value** – the optimal level of the decision variable (e.g., acres of wheat or corn).
- **Reduced Cost** – for nonbasic variables (value = 0), how much the objective coefficient must improve before the variable would enter the solution. Zero if the variable is positive in the solution.
- **Objective Coefficient** – the profit (or cost) per unit used in the objective function.
- **Allowable Increase / Decrease** – the range over which the objective coefficient can change without altering the current optimal basis (solution structure).

Interpretation of results

- Corn: optimal solution plants 450 acres of corn. Reduced cost = 0 because corn is already in the solution
- Wheat: optimal solution plants 0 acres of wheat. Reduced cost = -25 means a \$25/acre increase (from 200 to 225) reaches the alternative-optima threshold; an increase beyond \$25 makes wheat enter the solution.

Ranges:

- Wheat: profit can increase by 25 before the current corner ceases to be uniquely optimal.
- Corn: profit can fall as much as 33.3 (300 $\rightarrow$ 266.7) before solution changes.
- Interpretation: Corn dominates under current prices. Wheat only becomes attractive if its relative profit improves substantially.

3.8.2 Excel Bottom Panel: Constraints

Interpretation of columns:

- **Final Value** – the amount of the resource actually used at the solution.
- **Constraint R.H. Side** – the available amount of the resource (the right-hand side of the inequality).
- **Shadow Price** – the marginal value of relaxing the constraint (increase in objective if RHS increases by 1 unit), valid only within the allowable range.
- **Allowable Increase / Decrease** – the range over which the shadow price remains valid and the current basis stays optimal.

Interpretation of results:

- Land: only 450 acres used out of 500. Slack = 50 acres. Shadow price = 0 because land is not binding. You can increase land indefinitely without improving profit (since labor is the true bottleneck).
- Labor: fully used (1800/1800). Shadow price = 75 means each additional labor hour would increase profit by \$75, valid for up to +200 extra hours.

- Fertilizer: only 1350 units used of 2000. Slack = 650. Shadow price = 0 because it's not binding.

Economic Takeaways for Discussion

- Which resource is scarce? Labor.
 - What is the operation's reservation value for an additional hour of labor capacity? \$75 per hour, before paying any cost of obtaining that capacity.
 - Why does wheat drop out of the solution? Because relative to corn it uses too much labor per profit dollar.
 - Policy thought experiment: If labor availability were increased by 200 hours, profit would rise by \$15,000 (200×75).
-

3.9 Economic Interpretation

- **Shadow prices:** marginal value of resources.
 - **Allowable ranges:** robustness of those shadow prices.
 - **Managerial use:**
 - Identify which resources are most binding.
 - Assess which profit coefficients are critical.
 - Evaluate new technologies or policy changes.
-

3.10 Summary

- Sensitivity analysis extends LP results beyond a single point solution.
- Provides insight into:
 - Resource valuation (RHS changes),
 - Profit robustness (objective changes),
 - Technology shifts (coefficient changes).

4 Solve an LP in R

Work through R_lp_example.pdf

5 Duality in Linear Programming

5.1 Review

- Sensitivity analysis evaluates robustness of LP solutions to parameter changes.
 - Ranging analysis indicate how much parameters can change before the solution structure shifts.
- Shadow prices measure the marginal value of resources.
- Excel Solver provides a convenient way to solve LPs and get sensitivity reports.

5.2 Wheat–Corn primal and its dual

Return to the farmer’s profit-maximization problem. We call this original problem the **primal**; its **dual** will describe the value of the farm’s resources.

Primal (wheat–corn):

$$\begin{aligned} \max \quad & 200x_W + 300x_C \\ \text{s.t.} \quad & x_W + x_C \leq 500 \quad (\text{land}) \\ & 3x_W + 4x_C \leq 1800 \quad (\text{labor}) \\ & 4x_W + 3x_C \leq 2000 \quad (\text{fertilizer}) \\ & x_W, x_C \geq 0. \end{aligned}$$

5.2.1 An alternative to farming: offer the resources to someone else

Suppose another farmer offers to pay for the use of our land, labor capacity, and fertilizer for the production period. Our farmer can either use these resources to grow crops or receive payment for making them available to the buyer.

Let the buyer’s per-unit offers be:

- y_1 : dollars per acre of land;
- y_2 : dollars per hour of labor capacity;
- y_3 : dollars per unit of fertilizer.

The buyer wants to acquire the available resources at the lowest total payment:

$$\min_{y_1, y_2, y_3 \geq 0} 500y_1 + 1800y_2 + 2000y_3.$$

Without any restrictions, the buyer would offer zero. But the farmer has an alternative: use the resources to earn profit from wheat or corn. We therefore require the offers to **cover the production value of the resource bundle used by each crop**.

5.2.2 What makes an offer adequate?

One acre of wheat requires one acre of land, three labor hours, and four fertilizer units. The buyer’s payment for that bundle would be

$$y_1 + 3y_2 + 4y_3.$$

The farmer could use the same bundle to earn \$200 growing wheat. To cover that opportunity, the offers must satisfy

$$y_1 + 3y_2 + 4y_3 \geq 200.$$

Likewise, one acre of corn uses one acre of land, four labor hours, and three fertilizer units, and earns \$300. Thus,

$$y_1 + 4y_2 + 3y_3 \geq 300.$$

These conditions ensure that neither crop earns more than the value assigned to the resources it uses. Because production is linear, they also cover any feasible combination of wheat and corn.

The resulting dual is

$$\begin{array}{ll} \min & 500y_1 + 1800y_2 + 2000y_3 \\ \text{s.t.} & y_1 + 3y_2 + 4y_3 \geq 200 \quad (\text{wheat}) \\ & y_1 + 4y_2 + 3y_3 \geq 300 \quad (\text{corn}) \\ & y_1, y_2, y_3 \geq 0. \end{array}$$

The primal chooses **crop acreages** to maximize profit. The dual chooses **resource valuations** to minimize the total offer while covering every crop's production opportunity.

5.2.3 Interpreting the optimal offers

Recall that the optimal planting plan is 450 acres of corn and no wheat. It earns \$135,000, uses all 1,800 labor hours, and leaves 50 acres of land and 650 fertilizer units unused.

Consider the offers

$$y_1 = 0, \quad y_2 = 75, \quad y_3 = 0.$$

They satisfy both crop constraints:

$$\begin{array}{ll} \text{Wheat:} & 0 + 3(75) + 4(0) = 225 \geq 200, \\ \text{Corn:} & 0 + 4(75) + 3(0) = 300. \end{array}$$

The total offer is

$$500(0) + 1800(75) + 2000(0) = \$135,000.$$

This equals the profit from the best planting plan. The farmer is indifferent, within the model, between farming and accepting this payment. No feasible dual offer can be lower than the return from a feasible planting plan, so this equality establishes optimality.

These optimal offers are **shadow prices**: they measure the marginal value of additional resource availability to this farm.

- **Land:** $y_1 = 0$. More land alone cannot raise profit because some land already remains unused.
- **Labor:** $y_2 = 75$. Another labor hour permits another quarter-acre of corn, adding \$75 to profit before paying for the extra hour.
- **Fertilizer:** $y_3 = 0$. More fertilizer alone cannot raise profit because fertilizer already remains unused.

These are implicit values generated by the model's production opportunities and resource availability. They need not equal observed land rents, wages, or fertilizer prices. The \$75 labor value applies to the next 200 additional hours; beyond that, land prevents further expansion.

5.2.4 Why corn is grown and wheat is not

At the optimal resource valuations, corn's \$300 return exactly covers its resource opportunity cost. Wheat earns \$200 but uses resources valued at \$225.

The difference has a concrete interpretation: wheat's three labor hours could support three-quarters of an acre of corn, earning \$225. Growing an acre of wheat therefore sacrifices \$25 of profit.

This illustrates complementary slackness: a crop grown in positive quantity must exactly cover its resource opportunity cost; a crop whose resource opportunity cost strictly exceeds its return is excluded from the optimal plan.

Let's verify in excel

5.3 From the farm example to the general form

We constructed the dual by asking what the farm's resources must be worth. Now identify the mathematical structure that lets us use the same reasoning for other linear programs.

In the farm example, three resource constraints became three resource-price variables, and two crop variables became two valuation constraints. The resource endowments became the dual objective coefficients, while the crop returns became the dual right-hand sides. Each crop's column of resource requirements became a row in the dual.

Write x for activity levels, c for their per-unit returns, b for resource availability, and A for the matrix of resource requirements. For the wheat–corn example,

$$x = \begin{pmatrix} x_W \\ x_C \end{pmatrix}, \quad c = \begin{pmatrix} 200 \\ 300 \end{pmatrix}, \quad b = \begin{pmatrix} 500 \\ 1800 \\ 2000 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 3 & 4 \\ 4 & 3 \end{pmatrix}.$$

5.3.1 Canonical primal–dual pair

Primal: choose activity levels to maximize profit.

$$\begin{aligned} \max_{x \geq 0} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b. \end{aligned}$$

Dual: choose resource valuations to minimize total resource value while covering each activity's return.

$$\begin{aligned} \min_{y \geq 0} \quad & b^\top y \\ \text{s.t.} \quad & A^\top y \geq c. \end{aligned}$$

5.3.2 Constructing the dual in this form

For a maximization primal with $Ax \leq b$ and $x \geq 0$:

- Introduce one nonnegative dual variable for each resource constraint.
- Minimize $b^\top y$: available quantities times their resource valuations.
- Add one constraint per activity: its resource bundle must be valued at least as highly as its return. These constraints are $A^\top y \geq c$.

The transpose $A^\top$ puts each activity's resource requirements into one dual constraint. Nonnegative resource prices reflect that additional availability cannot reduce maximum profit. The $\geq$ constraints ensure that resource valuations cover production opportunities.

This gives a general foundation for the shadow prices we studied in sensitivity analysis. The following theorems explain why feasible resource valuations bound achievable profit, why the optimal values coincide, and how resource prices relate to the optimal activity mix.

5.4 Duality theorems

5.4.1 Weak duality

Suppose the buyer offers $y_1 = 0$, $y_2 = 100$, and $y_3 = 0$: nothing for land or fertilizer and \$100 per labor hour. These offers cover both crop opportunities:

$$\begin{aligned}\text{Wheat: } & 0 + 3(100) + 4(0) = 300 \geq 200, \\ \text{Corn: } & 0 + 4(100) + 3(0) = 400 \geq 300.\end{aligned}$$

The total offer is $500(0) + 1800(100) + 2000(0) = \$180,000$. Does this mean the farm can earn \$180,000? No. It means that **no feasible planting plan can earn more than that**. These resource valuations put a ceiling on profit, but the ceiling may be loose.

For any primal-feasible x and dual-feasible y , **weak duality** states that

$$c^\top x \leq b^\top y.$$

The economic reasoning is a chain of three quantities:

$$\underbrace{c^\top x}_{\text{crop profit}} \leq \underbrace{y^\top Ax}_{\text{value of resources used}} \leq \underbrace{y^\top b}_{\text{value of resources available}}.$$

The first inequality holds because each crop's return is no greater than the valuation of its resource bundle. Multiplying by acres planted and summing preserves that relationship. The second holds because the farmer cannot use more resources than are available: at nonnegative prices, the resources used cannot be worth more than the entire endowment.

Derivation step by step:

- From the primal constraints: $Ax \leq b$.
- From the dual constraints: $A^\top y \geq c$.
- Premultiply the dual constraint by $x^\top$, with $x \geq 0$: $x^\top A^\top y \geq x^\top c$. Note that $x^\top A^\top y = (Ax)^\top y$.
- Transpose the primal inequality and postmultiply by $y \geq 0$: $(Ax)^\top y \leq b^\top y$.
- Putting the two together gives $c^\top x \leq x^\top A^\top y = (Ax)^\top y \leq b^\top y$.
- If the primal is a minimization problem, the inequality is reversed: $c^\top x \geq b^\top y$.

For our feasible planting plan of 450 acres of corn, the \$100 labor valuation gives

$$\underbrace{135,000}_{\text{crop profit}} \leq \underbrace{180,000}_{\text{value of resources used}} \leq \underbrace{180,000}_{\text{value of resources available}}.$$

The gap is \$45,000. Feasibility alone does not imply that either the planting plan or the resource valuation is optimal.

5.4.2 Strong duality

The \$180,000 valuation covers the farmer's opportunities, but the buyer can do better. Lowering the labor offer to \$75 still covers both crops and reduces the total offer to \$135,000:

$$\begin{aligned}\text{Wheat: } & 0 + 3(75) + 4(0) = 225 \geq 200, \\ \text{Corn: } & 0 + 4(75) + 3(0) = 300, \\ \text{Total offer: } & 500(0) + 1800(75) + 2000(0) = 135,000.\end{aligned}$$

Meanwhile, the farmer can actually earn \$135,000 by planting 450 acres of corn. We now have a feasible planting plan earning \$135,000 and a feasible resource valuation ruling out anything higher. **There is no room left for either side to improve.**

Strong duality states that if either problem has a finite optimal solution, the other also has an optimal solution, and their objective values coincide:

$$\max_{x \geq 0, Ax \leq b} c^\top x = \min_{y \geq 0, A^\top y \geq c} b^\top y.$$

Equivalently, $c^\top x^* = b^\top y^*$. The tightest resource valuation exactly matches the best production plan. The two problems arrive at the same value by choosing different things: acres in the primal and resource prices in the dual.

5.4.3 Where does the gap come from?

Subtract the profit from the total resource valuation, then add and subtract the value of resources used:

$$\begin{aligned}b^\top y - c^\top x &= b^\top y - y^\top Ax + y^\top Ax - c^\top x \\ &= y^\top (b - Ax) + x^\top (A^\top y - c).\end{aligned}$$

There are two possible sources of a gap:

- $y^\top (b - Ax)$: **value assigned to unused resources.**
- $x^\top (A^\top y - c)$: **resource valuation above returns on crops grown.**

For feasible x and y , both terms are nonnegative. At optimality, strong duality says their sum is zero, so both must be zero.

In our optimal wheat–corn solution, unused land and fertilizer create no gap because their shadow prices are zero. Wheat's resource valuation exceeds its return, but that creates no gap because we grow no wheat. Corn is grown, and its return exactly equals its resource valuation.

This is the economic content of **complementary slackness**: neither source of a gap remains at the optimum.

5.4.4 Complementary slackness conditions

The **complementary slackness conditions** link the primal and dual solutions. They provide the bridge between activity levels and resource prices:

- For each **primal constraint** $a_i^\top x \leq b_i$ with dual variable $y_i \geq 0$:

$$y_i (b_i - a_i^\top x) = 0$$

This means that either:

- The constraint is **binding** ($a_i^\top x = b_i$) and then $y_i \geq 0$,
- Or the constraint is **slack** ($a_i^\top x < b_i$) and then $y_i = 0$.

- For each **primal variable** $x_j \geq 0$ with dual inequality $a_j^\top y \geq c_j$:

$$x_j (a_j^\top y - c_j) = 0$$

This means that either:

- The activity is **produced** ($x_j > 0$) and then its dual inequality binds exactly ($a_j^\top y = c_j$),
- Or the activity is **not produced** ($x_j = 0$) and then the dual inequality can be slack ($a_j^\top y > c_j$).

5.5 Summary

- Every LP has a corresponding dual LP.
- Dual variables represent shadow prices of primal constraints.
- Weak duality: primal objective $\leq$ dual objective for any feasible solutions.
- Strong duality: at optimality, primal objective = dual objective.
- Complementary slackness links primal and dual solutions, indicating which constraints and variables are binding.